

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (10 points) Find the Maclaurin series and its radius of convergence for the following functions.

(a) $f(x) = xe^x$

$$e^x = \sum_{n=0}^{+\infty} \frac{x^n}{n!}$$

$$\Rightarrow xe^x = x \sum_{n=0}^{+\infty} \frac{x^n}{n!} = \sum_{n=0}^{+\infty} \frac{x^{n+1}}{n!}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{x^{n+2}}{(n+1)!} \cdot \frac{n!}{x^{n+1}} \right| = \left| \frac{x}{n+1} \right| \rightarrow 0 \text{ as } n \rightarrow +\infty$$

$\Rightarrow$ the radius of convergence $R = +\infty$.

(b) $f(x) = \sin(2x)$

$$\sin x = \sum_{n=0}^{+\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\Rightarrow \sin 2x = \sum_{n=0}^{+\infty} \frac{(-1)^n (2x)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{+\infty} \frac{(-1)^n 2^{2n+1} x^{2n+1}}{(2n+1)!}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{(-1)^{n+1} 2^{2n+3} x^{2n+3}}{(2n+3)!} \cdot \frac{(2n+1)!}{(-1)^n 2^{2n+1} x^{2n+1}} \right| = \left| \frac{2^2 x^2}{(2n+2)(2n+3)} \right| \rightarrow 0 \text{ as } n \rightarrow +\infty$$

$\Rightarrow R = +\infty$

(c) $f(x) = (1-x)^{-2}$

(Hint: $(1+x)^k = \sum_{n=0}^{+\infty} \frac{k(k-1)\cdots(k-n+1)}{n!} x^n$)

$$\begin{aligned} (1-x)^{-2} &= \sum_{n=0}^{+\infty} \frac{(-2)(-3)(-4)\cdots(-2-n+1)}{n!} (-x)^n = \sum_{n=0}^{+\infty} \frac{(-1)^n 2 \cdot 3 \cdot 4 \cdots (n+1)}{n!} x^n \\ &= 1 + \frac{2}{1!}x + \frac{2 \cdot 3}{2!}x^2 + \frac{2 \cdot 3 \cdot 4}{3!}x^3 + \frac{2 \cdot 3 \cdot 4 \cdot 5}{4!}x^4 + \cdots \\ &= 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \cdots \\ &= \sum_{n=0}^{+\infty} (n+1)x^n \end{aligned}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{(n+2)x^{n+1}}{n x^n} \right| = \left| \frac{(n+2)}{n} x \right| \rightarrow |x| \text{ as } n \rightarrow +\infty$$

$\Rightarrow |x| < 1 \Rightarrow R = 1$.